

FLEXHAND · DESIGN DECISIONS REGISTER

FlexHand design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flexhand		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, FXH-DDR-001 to FXH-DDR-003 and the build plan work

FlexHand design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	First clinical co-design partner	A stroke unit occupational therapy team; a community rehabilitation service; a university rehabilitation lab	None yet	Nothing in the TRL 3 build; the therapist reviews in items 4 and 5 wait on it	FXH-DDR-001, O1
2	Accept the design-for-construction changes	(a) accept C1 to C14 as made; (b) ask for changes to any of them	(a)	The whole build plan	FXH-DDR-003, Table 1
3	Confirm the pull-out release plate as the tool-free release, in place of the concept's undefined lever	(a) pull-out plate, with pull force and release time measured at TRL 4 against R9's 10 s; (b) a cam lever that drives the same plate (more parts, lower pull force)	(a); move to (b) only if the measured pull is too high for a care partner	Release plate, cover and pucks (build plan 3.15 to 3.17, step 15)	FXH-DDR-003, A1
4	The cuffs need the fingers held slightly spread by the glove; spastic fingers may resist it	(a) accept, and have the co-design therapist check it; (b) single-saddle cuffs that fit fingers at rest, with higher pressure (R11 at risk on the little finger)	(a)	Finger cuffs and thimbles (build plan 3.12, 3.13)	FXH-DDR-003, A2
5	Forearm pack about 723 g against R7's 450 g target (639 g in the concept)	(a) keep the target under therapist review as already decided (N4 (c), waiting on item 1) and weigh at TRL 4; (b) look for mass now, starting with the savings below	(a), trying the mass savings below before TRL 4	Pack base, anchor block, fixings	FXH-DDR-003, A3; FXH-DDR-002, N4

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The gearbox face's two M3 holes: spacing (17 mm assumed), how deep a screw may go, and the shaft length (12.5 mm assumed)	They set the bulkhead holes, the 6 mm screw length and how far the spool hub reaches onto the shaft	FXH-DDR-003, C3, C4
2	The motor length with its encoder (71 mm assumed) and where the encoder cable leaves	Sets the 6 mm cable gap behind the motors	FXH-DDR-003, C5
3	The stop button's body depth (18 mm assumed), bezel and nut	Its body sits between the cells and the motors	FXH-DDR-003, C5
4	The sizes of the controller, driver, charger, battery board and regulator modules, and the charger's USB-C port position	They are laid out on the tray, and the port must meet the rear wall opening	FXH-DDR-003, C5
5	The cells' length with any protection fitted (65 mm assumed)	The saddles and tray posts leave 0.5 mm at each end	FXH-DDR-003, C5
6	The sheath housing's outside diameter (4 mm) and its ferrules' (about 5 mm)	They set the puck, stop block and cover holes	FXH-DDR-003, C7, C11
7	The glove's size and stretch against the plate and spacer positions	Plates and spacer are stitched where the model puts them	FXH-DDR-003, C11, C12
8	The breakaway couplings' pull-out force (about 60 N) and the stop-bead settings	Set and recorded at TRL 4	FXH-DDR-002, N3, N4

■ Value engineering

Value-engineering target: USD 500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 303.90 (USD 196.10 under the target), from `bom/bom.csv` [J1]. Main cost drivers and savings worth trying:

- **Gearmotors, USD 107.90 (36 %).** The largest line and 214 g of the pack's mass. Worth pricing a lighter 20 to 25 mm gearmotor class with encoder that still carries 0.49 N·m peak; any change reopens R2 and the torque calculation.
- **Battery, stop button and electronic modules, about USD 68.** Standard modules; little to save in cost, but a single dual-channel driver board would save one board, its wiring and a few grams.
- **Anchor block and its hardware, USD 18 plus fixings.** The printed parts are 71 g. A thinner-walled block with the channels as separate tubes, or printing it in a lighter material, is worth trying for mass.
- **Fixing kit, USD 10, 27 g.** Plastic or aluminium screws where loads are small (lid, tray, cover) would save about 8 g.
- **Pack base, 105 g.** Lower infill and thinner ribs could save 10 to 20 g; to check against the motor torque reaction.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D8: 25 mm 227:1 gearmotors on the forearm, two motors, passive motion only, widest anchor cuffs with a soft liner, 30 N design load, passive thumb, tendons in Bowden sheaths, 2S pack	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	FXH-DDR-001
2026-09-25	TRL 4 on hold for this repo	Amish, same instruction	<code>project.yaml</code> (trl_target), FXH-DDR-001
2026-09-25	Items N1 to N5: R2 restated for mild tone, fingertip thimbles, balance pulleys with stop beads, ball-detent couplings and a perforated cuff, 4 s lower stroke limit at design load	Amish: "i accept all your recommendations, go with them across all repos."	FXH-DDR-002
2026-09-26	FlexHand in the first batch of product renders	Amish chose the repo (no words recorded)	<code>docs/REVIEW.md</code> , session of 2026-09-26
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	FXH-DDR-003 (changes made under this instruction, open for review: item 2)
2026-09-30	Outstanding decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	This register
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering